

VECTOR BUNDLES ON FARGUES-FONTAINE CURVES

KIRAN S. KEDLAYA

Definition 1. A field F is *formally real* if -1 is not a sum of squares in F . A field F is *real closed* if it is formally real but has no proper finite extension which is formally real.

Lemma 2. Let F be a field of characteristic $p > 0$. Let $a \in F$ be an element which is not a p -th power. Then for every positive integer n , the polynomial $x^{p^n} - a$ is irreducible over F .

Lemma 3. Let F be a field which admits a finite algebraically closed extension. Then F is perfect.

Theorem 4. Let F be a field which admits a finite algebraically closed extension. Then F is algebraically closed or real closed.

1. ABSTRACT COMPLETE CURVES AND VECTOR BUNDLES

Definition 5. By an *abstract complete curve* we mean a connected, separated, regular 1-dimensional Noetherian scheme X , together with a *degree* function from closed points of X to positive integers with the property that the induced linear map $\deg: \text{Div}(X) \rightarrow \mathbb{Z}$ is zero on all principal divisors.

Definition 6. Since the degree map $\text{Div}(X) \rightarrow \mathbb{Z}$ vanishes on principal divisors, it induces a map $\text{Pic}(X) \rightarrow \mathbb{Z}$ which we use to define a *degree* homomorphism $\deg: K_0(X) \rightarrow \mathbb{Z}$.

Definition 7. Let $F \subseteq G$ be an inclusion of vector bundles on X . Then there is a unique intermediate bundle F' such that $\text{rank}(F) = \text{rank}(F')$ and G/F' is a vector bundle, called the *saturation* of F in G . We then have $\deg(F) \leq \deg(F')$ with equality if and only if $F' = F$.

Lemma 8. Every vector bundle F on X admits a filtration of the form

$$0 = F_0 \subset \cdots \subset F_n = F$$

in which F_i/F_{i-1} is a vector bundle of rank 1 for $i = 1, \dots, n$.

Proof. We proceed by induction on $\text{rank}(F)$, the case of rank 0 serving as a trivial base case. For the induction step, use the ampleness of $\mathcal{O}_X(1)$ to construct a surjection $\mathcal{O}_X^{\oplus d} \rightarrow F(m)$ for some positive integers d and m . In particular, there exists a nonzero (hence injective) map $\mathcal{O}_X(m) \rightarrow F$ for some integer m ; let F_1 be the saturation of the image of this map. We then apply the induction hypothesis to F/F_1 to conclude. $\square$

Definition 9. Let F be a nonzero vector bundle on X . We define the *slope* of F as

$$\mu(F) := \frac{\deg(F)}{\text{rank}(F)}.$$

The bundle F is *stable* if there is no nonzero proper subbundle G of F with $\mu(G) \geq \mu(F)$. The bundle F is *semistable* if there is no nonzero subbundle G of F with $\mu(G) > \mu(F)$. The bundle F is *polystable* if it is a direct sum of stable vector bundles, all of the same slope.

Lemma 10. *Let F, G be two stable vector bundles on X of the same slope μ . Then any nonzero morphism $f: F \rightarrow G$ is an isomorphism.*

Proof. Let $H \neq 0$ be the image of f . Let H' be the saturation of H in G . Then

$$\mu = \mu(F) \leq \mu(H) \leq \mu(H') \leq \mu(G) = \mu$$

so all of these inequalities must be equalities. Since G is stable this means that $H = H' = G$, and since F is stable the kernel of f must be zero. $\square$

Proposition 11. *Every vector bundle F on X admits a unique filtration*

$$0 = F_0 \subset F_1 \subset \cdots \subset F_l = F$$

in which each successive quotient F_i/F_{i-1} is a semistable vector bundle of some slope μ_i and $\mu_1 > \cdots > \mu_l$.

Proof. $\square$

Definition 12. For F a vector bundle on X , the filtration on F described in Theorem 11 is called the *Harder–Narasimhan filtration*, or *HN filtration*, of F . We define the multiset of *HN slopes* of F to consist of $\mu_i = \deg(F_i/F_{i-1})/\text{rank}(F_i/F_{i-1})$ with multiplicity $\text{rank}(F_i/F_{i-1})$.

Lemma 13. *Let F, G be two nonzero vector bundles on X . If $F \otimes G$ is semistable, then so are F and G .*

Proof. Let F_1, G_1 be the first steps in the respective HN filtrations of F and G ; then $\mu(F_1) \geq \mu(F)$, $\mu(G_1) \geq \mu(G)$. Then $F_1 \otimes G_1$ is a nonzero subbundle of $F \otimes G$ of slope $\mu(F_1) + \mu(G_1)$; since $F \otimes G$ is semistable, we must have

$$\mu(F_1) + \mu(G_1) \leq \mu(F \otimes G) = \mu(F) + \mu(G) \leq \mu(F_1) + \mu(G_1).$$

We conclude that $\mu(F_1) = \mu(F)$, $\mu(G_1) = \mu(G)$ and so F and G are semistable. $\square$

2. CLASSIFICATION OF VECTOR BUNDLES

Definition 14. Let $P := \bigoplus_{d \geq 0} P_d$ be a graded ring satisfying the following conditions.

- (i) The subring P_0 of P is a field and $P_1 \neq 0$.
- (ii) The multiplicative monoid

$$\left(\bigcup_{n=0}^{\infty} P_n \setminus \{0\} \right) / P_0^\times$$

is generated by $(P_1 \setminus \{0\})/P_0^\times$.

- (iii) For each nonzero $s \in P_1$, there exists a graded ring isomorphism

$$(1) \quad P/sP \cong P_0 \oplus TL_s[T]$$

for some field L_s containing P_0 .

Proposition 15. *The degree map $\deg: \text{Div}(X) \rightarrow \mathbb{Z}$ induces an isomorphism $\text{Pic}(X) \cong \mathbb{Z}$.*

Proof. The map $\text{Pic}(X) \rightarrow \mathbb{Z}$ carries $\mathcal{O}_X(n)$ to n , so we need only check injectivity. This follows from the fact that for any nonzero $s \in P_1$, $P[s^{-1}]_0$ is a principal ideal domain. $\square$

Lemma 16. *We have $H^i(X, \mathcal{O}_X(n)) = 0$ for all $n \geq 0$.*

Proof. For any $s, t \in P_1$ which are linearly independent over P_0 , by (1) we have $P_2 = P_1 s + P_1 t$. By induction on $m + n + k$ we deduce that $P_{m+n+k} = P_{m+k} s^n + P_{n+k} t^m$ for any $m, n > 0$ and $k \geq 0$, and hence

$$H^1(X, \mathcal{O}_X(n)) = \frac{P[(st)^{-1}]_n}{P[s^{-1}]_n + P[t^{-1}]_n} = \lim_{m \rightarrow \infty} \frac{P_{n+2m}}{P_{n+m} s^m + P_{n+m} t^m} = 0. \quad \square$$

Lemma 17. *Every vector bundle F on X admits a filtration of the form*

$$0 = F_0 \subset \cdots \subset F_n = F$$

in which each successive quotient F_i/F_{i-1} is isomorphic to $\mathcal{O}_X(m_i)$ for some $m_i \in \mathbb{Z}$.

Proof. Consider the filtration in Lemma 8. By Theorem 15, each successive quotient is isomorphic to $\mathcal{O}_X(m_i)$ for some $m_i \in \mathbb{Z}$. $\square$

Lemma 18. *Let F be a vector bundle on X admitting a filtration with successive quotients $\mathcal{O}_X(-1), \mathcal{O}_X^{\oplus n}, \mathcal{O}_X(1)$ for some $n \geq 0$. Then $H^0(X, F) \neq 0$.*

Proof. Choose $s \in P_1$ nonzero and let $x \in |X|$ be the unique closed point at which s vanishes. Define the bundle H to fit into the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{O}_X(-1) & \longrightarrow & F & \longrightarrow & F/\mathcal{O}_X(-1) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & \mathcal{O}_X & \longrightarrow & H & \longrightarrow & F/\mathcal{O}_X(-1) \longrightarrow 0 \end{array}$$

in which the left vertical arrow is injective with cokernel supported at x . Since $\text{Ext}_X^1(\mathcal{O}_X, \mathcal{O}_X) = H^1(X, \mathcal{O}_X)$ vanishes by Lemma 16, H admits a filtration with successive quotients $\mathcal{O}_X^{\oplus(n+1)}, \mathcal{O}_X(1)$.

We may assume that $H^0(X, H(-1)) = 0$, as otherwise H contains the subbundle $\mathcal{O}_X(1)$ whose intersection with F has degree ≥ 0 and hence by Proposition 15 is either $\mathcal{O}_X$ or $\mathcal{O}_X(1)$. Then $H^0(X, H)$ injects into $H^0(X, H/H(-1))$, which we reinterpret as the fiber H_x of H at x . Using the exact sequence

$$0 \rightarrow H^0(X, \mathcal{O}_X^{\oplus(n+1)}) \rightarrow H^0(X, H) \rightarrow H^0(X, \mathcal{O}_X(1)) \rightarrow H^1(X, \mathcal{O}_X^{\oplus n}) = 0$$

(where the last equality comes from theorem 16), we express the image of $H^0(X, H) \rightarrow H_x$ as $P_0 e_1 + \cdots + P_0 e_{n+1} + L_s e_{n+2}$ for some basis $e_1, \dots, e_{n+2}$ of H_x over L_s .

Meanwhile, the image of $H^0(X, F) \subset H^0(X, H)$ in H_x is the intersection of the image of $H^0(X, H)$ with the image of $F_x \rightarrow H_x$; the latter is a codimension 1 subspace of H_x and so can be written as the kernel of a nonzero L_s -linear functional $\lambda: H_x \rightarrow L_s$. Define $a \in P_0, b \in L_s$ by

$$(a, b) := \begin{cases} (0, 1) & \lambda(e_{n+2}) = 0 \\ (1, -\lambda(e_1)/\lambda(e_{n+2})) & \lambda(e_{n+2}) \neq 0; \end{cases}$$

then $ae_1 + be_{n+2}$ is the image in H_x of a nonzero element of $H^0(X, F)$. $\square$

Lemma 19. *With notation as in Lemma 8, we can further ensure that*

$$m_j \leq m_i + 1 \quad (1 \leq i < j \leq n).$$

Proof. Among filtrations as in Lemma 17, choose one to minimize $\sum_{i=1}^n (n-i)m_i$. We prove that this filtration has the desired property by induction on $j-i$.

We start with the base case $j = i+1$. Suppose by way of contradiction that this fails for some i . We then have an exact sequence

$$0 \rightarrow \mathcal{O}_X(m_i) \rightarrow F_{i+1}/F_{i-1} \rightarrow \mathcal{O}_X(m_{i+1}) \rightarrow 0.$$

By choosing an embedding $\mathcal{O}_X(m_i+2) \rightarrow \mathcal{O}_X(m_{i+1})$, we obtain a subbundle G of F_{i+1}/F_{i-1} and an exact sequence

$$0 \rightarrow \mathcal{O}_X(m_i) \rightarrow G \rightarrow \mathcal{O}_X(m_i+2) \rightarrow 0.$$

By Lemma 18, G admits $\mathcal{O}_X(m_i+1)$ as a (not necessarily saturated) subbundle. We can thus replace the terms m_i, m_{i+1} in our original sequence by new terms m'_i, m'_{i+1} with $m'_i > m_i$ and $m'_i + m'_{i+1} = m_i + m_{i+1}$, contradicting minimality.

We next proceed to the induction step $j-i > 1$. Suppose by way of contradiction that this fails for some i . By the induction hypothesis, we have $m_{j-1} \leq m_i+1$ and $m_j \leq m_{j-1}+1$, so both of these must be equalities; in particular, $m_j = m_i+2$. For $k = i+1, \dots, j-1$, the induction hypothesis yields $m_i+1 \leq m_k \leq m_j-1$, so the sequence $(m_i, \dots, m_j)$ looks like $(m-1, m, \dots, m, m+1)$ for some m . That is, F_j/F_i admits a filtration with successive quotients $\mathcal{O}_X(m-1), \mathcal{O}_X(m)^{\oplus(j-i-1)}, \mathcal{O}_X(m+1)$. By Lemma 18 again, F_j/F_i admits $\mathcal{O}_X(m)$ as a (not necessarily saturated) subbundle, but again this would contradict minimality. $\square$

Proposition 20. *Every vector bundle on X which is semistable of slope 0 is trivial.*

Proof. Let F be a semistable vector bundle of degree 0 and rank n , and choose a filtration as in Lemma 19. Since $\sum_{i=1}^n m_i = \deg(F) = 0$, we have

$$0 \leq \max_i \{m_i\} \leq m_1 + 1$$

and so $m_1 \geq -1$; moreover, if $m_1 = -1$ we deduce that $\sum_{i=1}^n m_i < m_1 + \sum_{i=2}^n m_i \leq -1 + 0 < 0$, a contradiction. Hence $m_1 = 0$, and similar logic yields $m_n = 0$; by semistability we deduce that $m_1 = \dots = m_n = 0$. Since $\text{Ext}_X^1(\mathcal{O}_X, \mathcal{O}_X) = H^1(X, \mathcal{O}_X)$ vanishes by Lemma 16, it follows that F is trivial. $\square$

Lemma 21. *Let F be a vector bundle on X with all HN slopes in $[0, 1)$. Then for any closed point $x \in |X|$, for $j_x: x \rightarrow X$ the canonical inclusion, there exist a coherent sheaf G on x and an exact sequence*

$$0 \rightarrow \mathcal{O}_X^{\oplus \text{rank}(F)} \rightarrow F \rightarrow j_x^* G \rightarrow 0$$

of coherent sheaves on X .

Proof. We show that for $i = 1, \dots, \deg(F)$, we can find an exact sequence

$$(2) \quad 0 \rightarrow H_i \rightarrow F \rightarrow j_x^* G_i \rightarrow 0$$

in which H_i is a vector bundle of the same rank F with degree $\deg(F) - i$ and all HN slopes nonnegative (and hence in $[0, 1)$). For $i = \deg(F)$, H_i will then be semistable of degree 0, and hence trivial by Proposition 20.

In the case $i = 1$, it suffices to check the claim after replacing F with its maximum quotient having all HN slopes in $(0, 1)$; at this point, *every* exact sequence of the form (2), in which H_1 is a vector bundle of the same rank F with degree $\deg(F) - 1$, has the property that the HN slopes of H_1 are nonnegative. We achieve the general case by applying this argument repeatedly. $\square$

Corollary 22. *Let F be a nonzero vector bundle on X whose HN slopes are all nonnegative. Then F is generated by global sections and $H^1(X, F) = 0$.*

Proof. By replacing F with the first step of its HN filtration, we may reduce to the case where F is semistable. By twisting, we may further ensure that $\mu(F) \in [0, 1)$. We then deduce both claims from Lemma 16 and Lemma 21. $\square$

Lemma 23. *For any semistable vector bundle F on X , $F^\vee \otimes F$ is trivial.*

Proof. We immediately reduce to the case where F is stable with slope in $[0, 1)$. In this case, suppose by way of contradiction that $F^\vee \otimes F$ is not semistable, and let G be the first step of its HN filtration. Since $G \otimes G$ has slope $2\mu(G) > \mu(G)$, the composition map $(F^\vee \otimes F) \otimes (F^\vee \otimes F) \rightarrow F^\vee \otimes F$ restricts to zero on $G \otimes G$. By Corollary 22, G is generated by global sections; consequently, $G \otimes G$ is generated by nonzero global sections of the form $s_1 \otimes s_2$. Any such section corresponds to a pair of nonzero endomorphisms $F \rightarrow F$ which compose to zero; this contradicts Lemma 10. We deduce that $F^\vee \otimes F$ is semistable of slope 0, and hence trivial by Proposition 20. $\square$

Proposition 24. *The tensor product of any two semistable vector bundles on X is semistable.*

Proof. Let F, G be two semistable vector bundles on X . By Lemma 23,

$$(F^\vee \otimes G^\vee) \otimes (F \otimes G) \cong (F^\vee \otimes F) \otimes (G^\vee \otimes G)$$

is semistable. By Lemma 13, $F \otimes G$ must be semistable. $\square$

Lemma 25. *Any exact sequence of vector bundles on X of the form*

$$0 \rightarrow F_1 \rightarrow F \rightarrow F_2 \rightarrow 0,$$

in which every HN slope of F_1 is greater than or equal to every HN slope of F_2 , is split.

Proof. By Proposition 24, $F_2^\vee \otimes F_1$ has nonnegative HN slopes. Now $\text{Ext}_X^1(F_2, F_1) = H^1(X, F_2^\vee \otimes F_1)$ which vanishes by Corollary 22. $\square$

Proposition 26. *Every vector bundle on X splits (nonuniquely) as a direct sum of stable bundles. In particular, the HN filtration always splits and every semistable bundle is polystable (i.e., a direct sum of stable bundles of the same slope).*

Proof. Both assertions follow directly from Lemma 25. $\square$

Lemma 27. *For d a positive integer with $\dim_{P_0} P_1 > d$ and $c \in \mathbb{Z}$, there exists a semistable vector bundle on X of rank d and degree c .*

Proof. We may assume without loss of generality that $d > 1$ and $c \in \{1, \dots, d-1\}$. Choose $s \in P_1$ nonzero and let $x \in |X|$ be the unique closed point at which s vanishes. Set $F := \mathcal{O}_X^{\oplus d}$ and let $e_1, \dots, e_d$ be the images in F_x of the standard basis vectors of $H^0(X, F) \cong P_0^d$. The condition $\dim_{P_0} P_1 > d$ is equivalent to $\dim_{P_0} L_s \geq d$; consequently, we can choose elements $f_1, \dots, f_c \in L_s$ such that $1, f_1, \dots, f_c$ are linearly independent over P_0 . Let $V \subset F_x$ be the L_s -span of $e_1 - f_1 e_d, \dots, e_c - f_c e_d$ and construct the subbundle G of F so that we have an exact sequence

$$0 \rightarrow G \rightarrow F \rightarrow j_x^* V \rightarrow 0;$$

then $G^\vee$ is semistable of degree c . $\square$

Lemma 28. *For any $\lambda \in \mathbb{Q}$, there is at most one isomorphism class of stable vector bundles on X of slope λ .*

Proof. Let F, G be two such bundles. By Proposition 24, $F^\vee \otimes G$ is trivial. We can thus find a nonzero morphism $f: F \rightarrow G$, which must be an isomorphism by Lemma 10. $\square$

Theorem 29. *Set*

$$S := \begin{cases} \mathbb{Z} & \text{in the classical case,} \\ \frac{1}{2}\mathbb{Z} & \text{in the twistor case,} \\ \mathbb{Q} & \text{otherwise.} \end{cases}$$

Then the following statements hold.

- (a) *There exists exactly one isomorphism class of stable vector bundles $\mathcal{O}_X(\lambda)$ of slope λ if $\lambda \in S$ and none if $\lambda \notin S$.*
- (b) *For $\lambda \in S$ with denominator d in lower terms, $\text{rank}(\mathcal{O}_X(\lambda)) = d$.*
- (c) *The endomorphisms of $\mathcal{O}_X(\lambda)$ form a division algebra of degree d over P_0 .*
- (d) *For $\lambda, \lambda' \in S$, $\mathcal{O}_X(\lambda) \otimes \mathcal{O}_X(\lambda') \cong \mathcal{O}_X(\lambda + \lambda')^{\oplus m}$ for some positive integer m .*
- (d) *Every vector bundle on X splits (nonuniquely) as a direct sum of summands each of the form $\mathcal{O}_X(\lambda)$ for some $\lambda \in S$.*

Proof. If $\dim_{P_0} P_1 = \infty$, then we may deduce (a) and (b) from Lemma 27 and Lemma 28. Otherwise, by Theorem 4 we may argue as follows.

- In the classical case, $P = P_0[x, y]$ and $H^1(X, \mathcal{O}_X(-1)) = 0$. Hence any exact sequence of the form

$$0 \rightarrow \mathcal{O}_X \rightarrow F \rightarrow \mathcal{O}_X(1) \rightarrow 0$$

splits, as then does any filtration as in Lemma 19, proving (a). From this, (b) is obvious.

- In the twistor case, by identifying $P \otimes_{P_0} P_0(\sqrt{-1})$ with $P_0(\sqrt{-1})[x^2, xy, y^2]$ and applying the classical case, we see that $S \subseteq \frac{1}{2}\mathbb{Z}$. We deduce (a) and (b) by applying Lemma 27 to construct a stable bundle of rank 2 and degree 1, then applying Lemma 28 for uniqueness.

Given (a)–(b), (c) follows from Lemma 10 and Lemma 23; (d) is immediate from Proposition 24 and Proposition 26; and (e) is immediate from Proposition 26. $\square$